

MITUL MARIMUTHU

650-382-9253 | mitul6m16apply@gmail.com | mitulportfolio.vercel.app | linkedin.com/in/mitul-marimuthu

Education

University of California, Santa Barbara

Bachelor of Science in Computer Science, GPA: 3.95/4.0

Grad: June 2027

Santa Barbara, CA

Relevant coursework: Data Structures and Algorithms, Computer Architecture, Operating Systems, Artificial Intelligence, Computational Science, Formal Languages, Cryptography

Experience

Software Engineering Intern

Yope

Dec 2025 – March 2026

Remote

- Ideated, prototyped, and delivered 3 core app features, leading to a 30% increase in US download rates.
- Presented insights from 100+ user interviews, identifying and resolving 4 critical friction points, resulting in a \$15k increase in monthly recurring revenue.

Software Engineering Intern

smartQED

Jun 2025 – Sept 2025

Remote

- Automated client onboarding and 2 core services via an AI agent, cutting average service time by 30%.
- Implemented Retrieval-Augmented Generation (RAG) architecture to search through 200+ client cases, cutting costs for servicing 30 new clients by 40%.
- Researched 10 Agentic AI architectures and implemented proof of concepts with 95% accuracy, compiling findings in a detailed 20 page design document.

Undergraduate ML Researcher

UC Santa Barbara NLP Group

Apr 2026 – Present

Santa Barbara, CA

- Engineered a scalable AI training platform, enabling 20+ researchers to train research agents with higher reliability.
- Enhanced a multi-agent simulation environment, discovering leakage rates as high as 61.4% in state-of-the-art LLMs.

Leadership and Involvement

Technical Director

Association for Computing Machinery, UCSB

Jan 2025 – Present

Santa Barbara, CA

- Organized PokerBots, a school-wide poker AI Agent competition with 250 participants and 7 industry guest speakers.
- Developed an interface using Next.js and Supabase realtime for all contestants to compete, supporting 80+ bots and 300+ spectators for 4 hours.

Software Engineering Director

AgentsSB

Sept 2025 – Present

Santa Barbara, CA

- Directed a 6 person team to develop 4 AI-powered dashboards, leveraging agentic development environments to accelerate the product lifecycle by 50%.
- Established a curriculum centralized on Claude Code and Codex subagents and skills, enabling 60+ members to start deploying industry-level code.

Projects

MAGPIE Replication Report | *Python, Mistral, Next.js, Vercel*

Apr 2026 - May 2026

- Built a multi-agent LLM simulation framework from scratch, orchestrating 20+ high-stakes negotiation runs.
- Analyzed 400+ item-level leakage records, identifying reject-reason actions leak at 2.4x the rate of formal proposals, a finding not reported in the original paper.

Molecular Property Prediction Model | *Python, PyTorch, RDKit, Scikit-learn*

Apr 2026 - Present

- Implemented a graph neural network from scratch to predict molecular toxicity, combining cheminformatics (RDKit) with modern deep learning on graph-structured data.
- Trained on the OGB-MOLHIV benchmark, achieving 0.76 ROC-AUC with 0.62 F1 score on the minority class.
- Applied gradient attribution to visualize atom-level importance scores for model interpretability.

Technical Skills

Languages: Java, C/C++, Python, SQL, Javascript, Typescript, OCaml, R

Tools: MongoDB, Docker, Git, Supabase, PyTorch, HuggingFace

Frameworks: LangChain, LangGraph, Numpy, Pandas, React.js, Node.js, Next.js, CSS, Flask, Scikit-learn